

# TITLE : Synthesis & Gate-Level Simulation

## OBJECTIVE :

To synthesize a Verilog RTL design into a gate-level netlist using Icarus verilog on EDA Playground and perform gate-level simulation to verify that the synthesized circuit produces the same functionality as the original RTL design.

## Design Code:

```
module adder(  
    input [3:0] a,  
    input [3:0] b,  
    output [4:0] sum  
);  
  
assign sum = a + b;  
  
endmodule
```

## Testbench Code:

```
module testbench;  
  
    reg [3:0] a, b;  
    wire [4:0] sum;  
  
    adder uut(  
        .a(a),  
        .b(b),  
        .sum(sum)  
    );  
  
    initial begin  
        $dumpfile("dump.vcd");  
        $dumpvars(0,testbench);  
    end  
endmodule
```

```

a = 4'd5;
b = 4'd3;
#10;

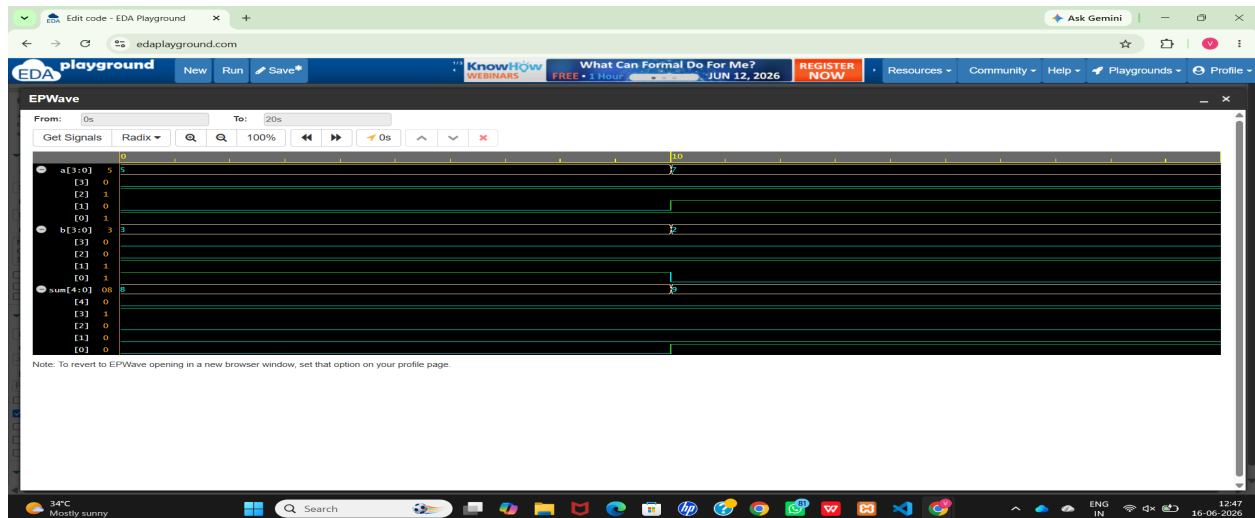
a = 4'd7;
b = 4'd2;
#10;

$finish;
end

endmodule

```

## Result :



## Conclusion :

Synthesis and Gate-Level Simulation are crucial steps in the digital VLSI design flow that bridge the gap between RTL design and physical hardware implementation.